ENGR 225 Homework 7 Questions

Assc. Prof. Clayton Byers

Assigned: 2 Nov 2026

Due: 10 Nov 2026 (11:59pm on Moodle)

Important notes before you begin:

- Note the due date is Tuesday before the second exam. There is some extra time to work on this assignment.
- On problem 2, the unit of *kip* means kilo-pounds.
- For problems 5, 6, and 7, your graphs should be as precise as possible, lining up critical location of the load with the internal shear and moment like was done in class. Indicate location where the functions change and give values at maximums, minimums, and zero-crossings.
- Problem 8 requires a bit of thinking when it comes to solving the equations and determining your unknowns. This is somewhat like problem 1 on HW 5 with the interconnection of equations and unknowns.

Problem 1

Determine the shear force and moment at points C and D. (**Ans:** $M_C = -857\,lb\cdot ft$, $V_C = -386\,lb$, $M_D = -600\,lb\cdot ft$, and $V_D = 300\,lb$.)

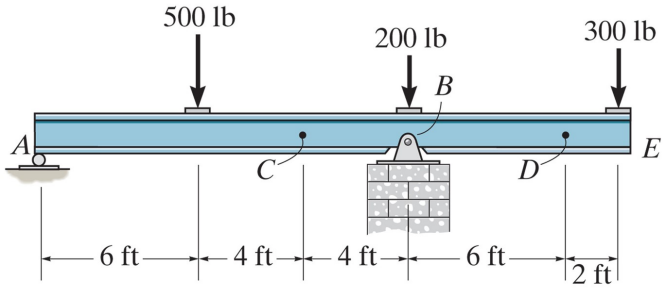


Figure for problem 1.

Problem 2

Determine the internal normal force, shear force, and moment in the beam at sections through points D and E. Assume the $5\,kip$ load is located just to the right of point D. (**Ans:** $M_D = 12\,kip \cdot ft$, $V_D = 3\,kip$, $N_D = 0$, $M_E = -20\,kip \cdot ft$, $V_E = -8\,kip$, and $N_E = 0$.)

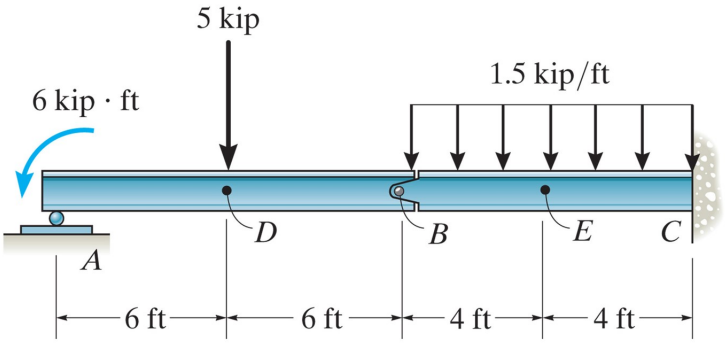


Figure for problem 2.

Problem 3

Determine the internal reactions at point C. (**Ans:** $M_C = 200\,N \cdot m$, $V_C = 200\,N$, $N_C = -1600\,N$)

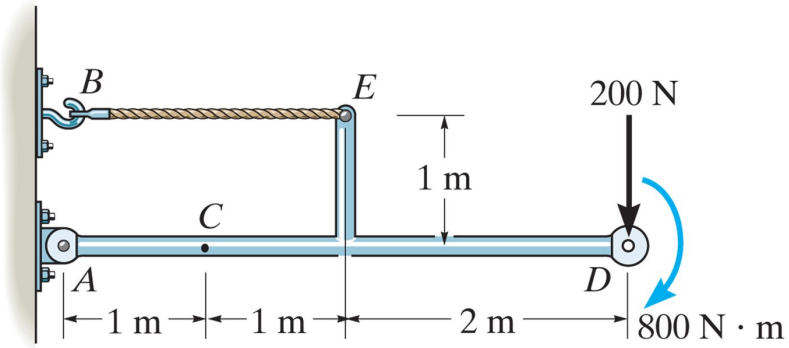


Figure for problem 3.

Problem 4

Determine the x, y, and z components of internal loading at a section passing through the point B in the pipe assembly. Neglect the weight of the pipe, and set $\vec{F}_1 = (100\hat{i} - 200\hat{j} - 300\hat{k})\,N$ and $\vec{F}_2 = (100\hat{i} + 500\hat{j})\,N$. (**Ans:** $F_x = -200\,N$, $F_y = -300\,N$, $F_z = 300\,N$, $M_x = 450\,N \cdot m$, $M_y = -300\,N \cdot m$, and $M_z = -150\,N \cdot m$)

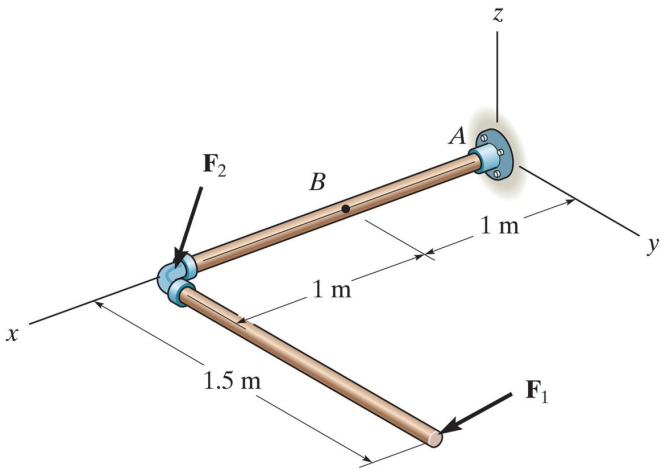


Figure for problem 4.

Problem 5

Draw the shear and moment diagrams for the beam with the given loading. Specify the equations for the shear and moment as a function of horizontal position. (**Ans:** $(0 \le x \le 12)$: $V(x) = -\frac{1}{6}x^2 + 48$, $M(x) = -\frac{1}{18}x^3 + 48x - 576$; $(12 \le x \le 24)$: $V(x) = \frac{1}{6}x^2 - 8x + 96$, $M(x) = \frac{1}{18}x^3 - 4x^2 + 96x - 768$)

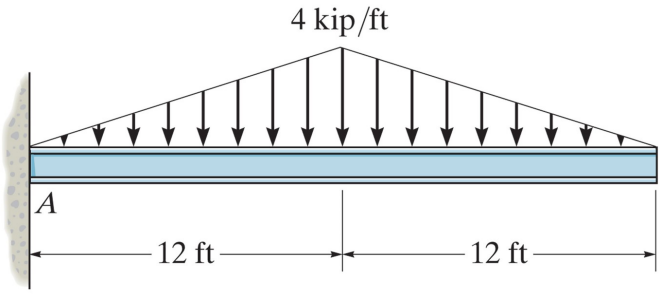


Figure for problem 5.

Problem 6

Draw the shear and moment diagrams for the beam with the given loading. Specify the equations for the shear and moment as a function of horizontal position. (**Ans:** $(0 \leq x \leq 6)$: $V(x) = -\frac{1}{4}x^2 + 1.5$, $M(x) = -\frac{1}{12}x^3 + 1.5x$; $(6 \leq x \leq 9)$: $V(x) = -2x + 18$, $M(x) = -x^2 + 18x - 81$)

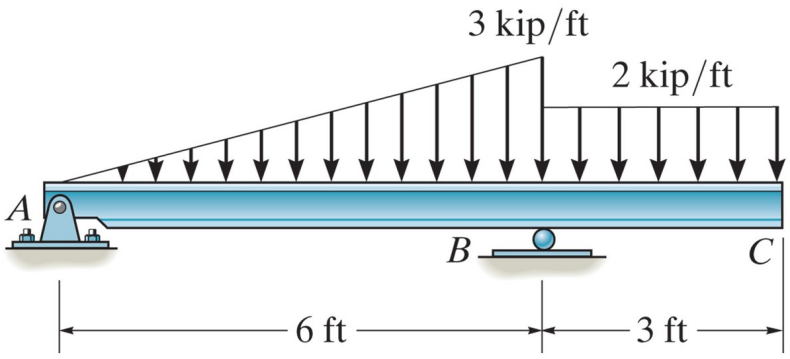


Figure for problem 6.

Problem 7

Draw the shear and moment diagrams for the beam with the given loading. Specify the equations for the shear and moment as a function of horizontal position. (**Ans:** $(0 \leq x \leq 6)$: $V(x) = -33.3x^2 + 500$, $M(x) = -11.1x^3 + 500x$; $(6 \leq x \leq 12)$: $V(x) = -33.3x^2 + 400x - 1900$, $M(x) = -11.1x^3 + 200x - 1900x + 7200$; $(12 \leq x \leq 16)$: $V(x) = 1500$, $M(x) = 1500x - 24000$)

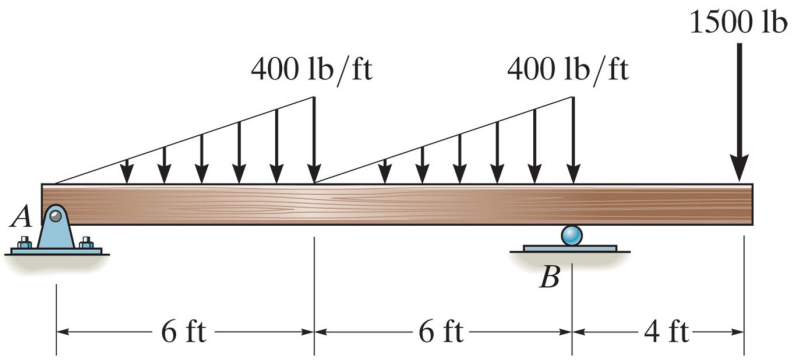


Figure for problem 7.

Problem 8

Determine the force P needed to hold the cable in the position as shown (so segment BC remains horizontal). With this, compute the sag y_B and the maximum tension in the cable. (**Ans:** $P = 0.8\,kN$, $y_B = 3.53\,m$, and $T_{max} = T_{DE} = 8.17\,kN$)

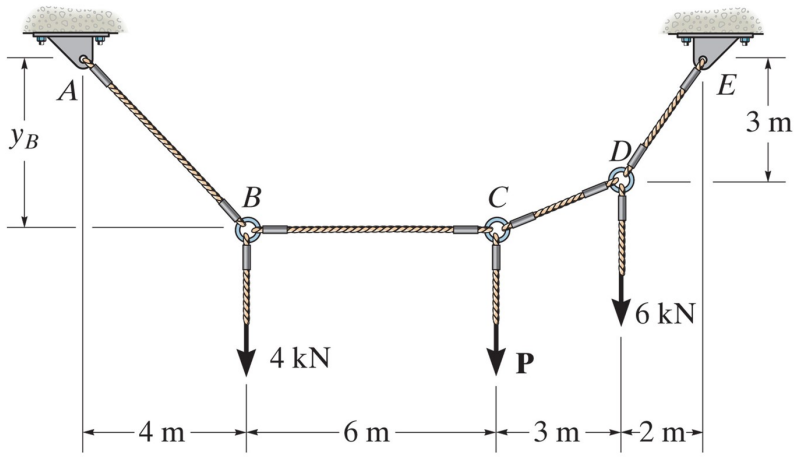


Figure for problem 8.